

GRAFICA DE FUNCIONES

TAREA

```
import numpy as np

import matplotlib.pyplot as plt

import sympy as sp

t_sym = sp.symbols('t')

A=1

x_sym= A*sp.exp(-
1/3*t_sym)*sp.cos(2*sp.pi*t_sym+sp.p
i/4)

v_sym=sp.diff(x_sym, t_sym)

a_sym=sp.diff(v_sym, t_sym)

x_func = sp.lambdify(t_sym, x_sym,
'numpy')

v_func = sp.lambdify(t_sym, v_sym,
'numpy')

a_func = sp.lambdify(t_sym, a_sym,
'numpy')

t = np.linspace(0, 6, 500)

fig, axs = plt.subplots(2, 2, figsize=(10,
6))

axs[0, 0].plot(t, x_func(t), color='red')
axs[0, 0].set_title("Posición")

axs[0, 1].plot(t, v_func(t), color='blue')
axs[0, 1].set_title("Velocidad")

axs[1, 0].plot(t, a_func(t), color='green')
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axs[1, 0].set_title("Aceleración")

plt.tight_layout()

plt.show()

from scipy.integrate import
cumulative_trapezoid, simpson

from ipywidgets import interact,
FloatSlider

t_full = np.linspace(0, 6, 500)

v = np.exp(-1/3 * t_full) * np.cos(2 *
np.pi * t_full + np.pi/4)

a = np.gradient(v, t_full)

def mostrar_integral(t_max):

    mask = t_full <= t_max

    t_sub = t_full[mask]

    v_sub = v[mask]

    resultado_area=simpson(y=v_sub,
x=t_sub)

    print(f"El área bajo la curva v(t) hasta
{t_max:.1f}s es: {resultado_area_v:.4f}")

    int_v_trap =
cumulative_trapezoid(v_sub, t_sub,
initial=0)
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plt.figure(figsize=(8, 4))

plt.plot(t_sub, int_v_trap,
label='Integral v(t) - Trapecio')


plt.text(0.1, np.max(int_v_trap)*0.8,
f"Área V: {resultado_area_v:.4f}",
        bbox=dict(facecolor='yellow',
alpha=0.5))
```

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plt.title(f"Integrales hasta t =
{t_max}s")

plt.legend()

plt.grid(True)

plt.show()

interact(mostrar_integral,
t_max=FloatSlider(value=3, min=0.1,
max=6, step=0.1));
```